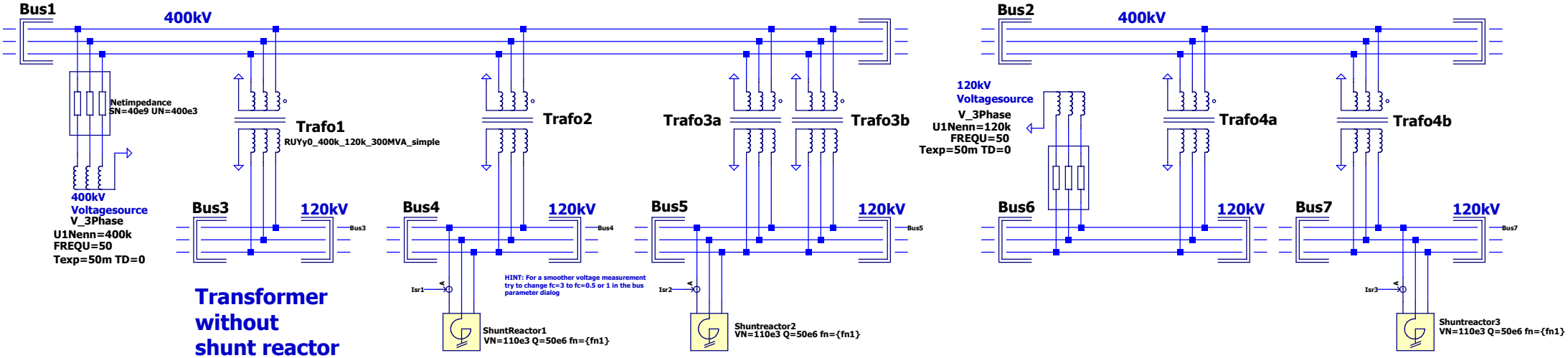
Transformer and Shunt Reactor TRAN-Analysis

```
.param tdv=1 tdt=1 fn1=50 omega_n={2*pi*fn1}
.options vntol=1m abstol=1u
```

.tran 0 4 0 0.1m



Transformer without shunt reactor

Transformer with shunt reactor

Two tranformers in parallel and shunt reactor

Two transformeres in series and shunt reactor

See the measurements of
Reactive power , voltage and current
of the three Shuntreactors
Activate schematic window
View->Spice Output log
Scroll to the end of the file

```
; Reactive powers
.meas TRAN Qsr_1 PP 3*Ix(ShuntReactor1:L1)*v(Bus4)/sqrt(2)/sqrt(3)/2 FROM 3.9 TO 4
.meas TRAN Qsr_2 PP 3*Ix(ShuntReactor2:L1)*v(Bus5)/sqrt(2)/sqrt(3)/2 FROM 3.9 TO 4
.meas TRAN Qsr_3 PP 3*Ix(ShuntReactor3:L1)*v(Bus7)/sqrt(2)/sqrt(3)/2 FROM 3.9 TO 4
; Voltages:
.meas TRAN Vsr_1  MAX v(Bus4)
.meas TRAN Vsr_2  MAX v(Bus5)
.meas TRAN Vsr_3  MAX v(Bus7)
; Currents
.meas TRAN Isr_1 MAX  Ix(ShuntReactor1:L1)/sqrt(2)
.meas TRAN Isr_2 MAX  Ix(ShuntReactor2:L1)/sqrt(2)
.meas TRAN Isr_3 MAX  Ix(ShuntReactor3:L1)/sqrt(2)
```

To see the simulation voltage results simply click on the lines
To see the currents click on the input pins of the device
The left buses measurement pins shows the line to line sinus voltages
The right buses measurement pins shows the line to line RMS voltages